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Density and Contour Plot Analysis using Iris Dataset

1. Objective

The objective of this project is to analyze data distribution using **density plots (KDE)** and **contour plots**, and to interpret patterns, clusters, and relationships in the Iris dataset.

2. Dataset Description

The **Iris dataset** is a well-known dataset in data science. It contains:

- 150 samples of iris flowers
- 3 species:
  - Setosa
  - Versicolor
  - Virginica
- 4 numerical features:
  - Sepal Length
  - Sepal Width
  - Petal Length
  - Petal Width

In this project, we mainly use:

- **Petal Length**
- **Petal Width**

3. Theory

🔹 Probability Density

Probability density describes how values of a continuous variable are distributed.

- Higher peaks → more data concentration
- Lower areas → fewer data points

🔹 Histogram vs Density Plot

Histogram	Density Plot
Uses bars	Smooth curve
Depends on bin size	Continuous
Less precise	More accurate

🔹 Density Plot (KDE)

- KDE (Kernel Density Estimation) creates a smooth curve
- Helps understand distribution shape clearly

🔹 Contour Plot

- Represents **2D density**
- Lines connect points with equal density
- High-density regions = clusters

Step 1: Import Libraries

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
```

Step 2: Load Dataset

```
iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)

df['species'] = iris.target
```

Output:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Step 3: Convert Species Names

```
df['species'] = df['species'].map({
   0: 'setosa',
   1: 'versicolor',
   2: 'virginica'
})
df.head()
```

Step 4: Single Density Plot

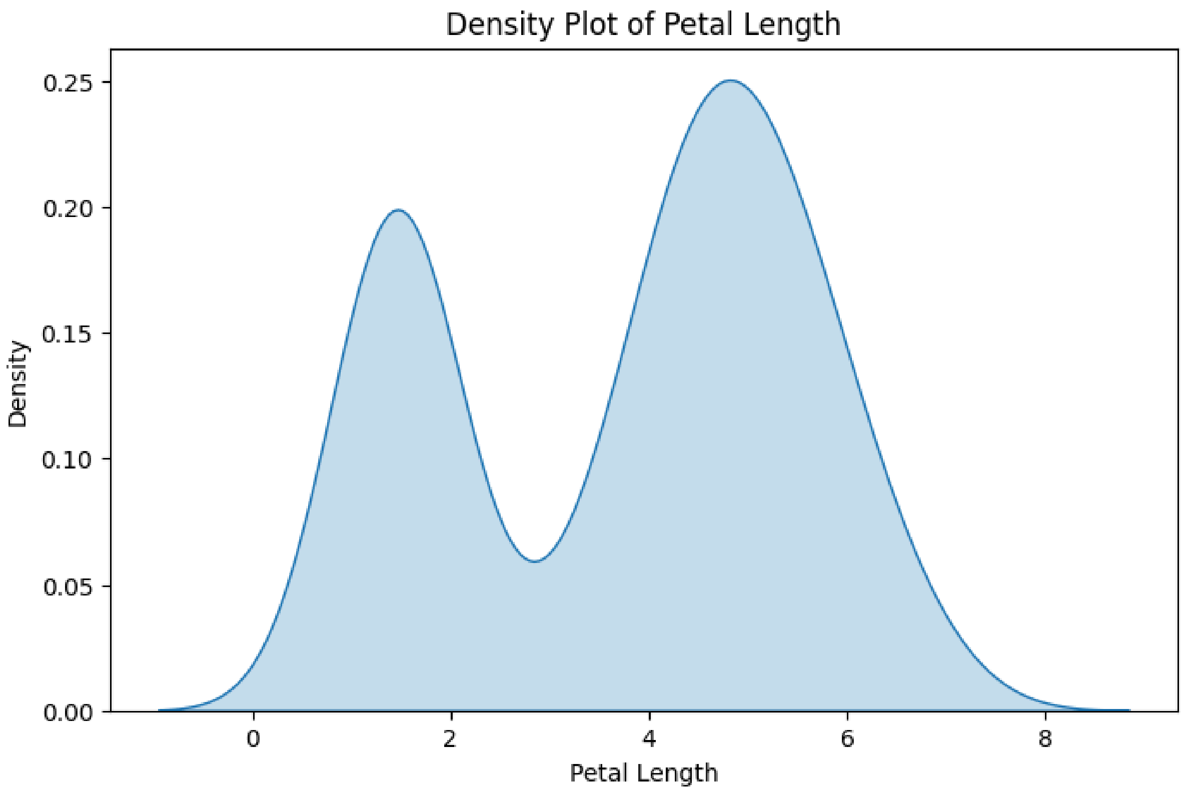
```
plt.figure(figsize=(8,5))

sns.kdeplot(data=df, x='petal length (cm)', fill=True)

plt.title("Density Plot of Petal Length")
plt.xlabel("Petal Length")
plt.ylabel("Density")

plt.show()
```

Output:



Step 5: Multiple Density Plot (Overlay)

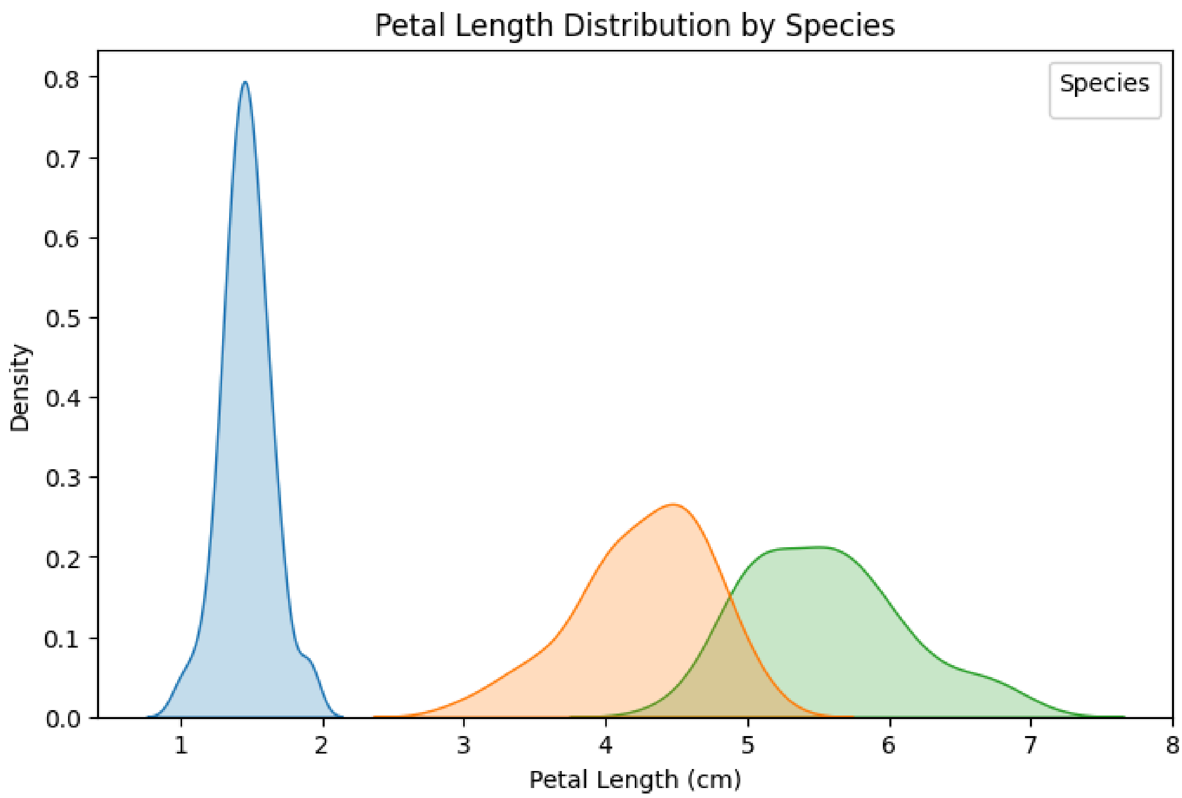
```
plt.figure(figsize=(8,5))

sns.kdeplot(
  data=df,
  x='petal length (cm)',
  hue='species',
  fill=True
)

plt.title("Petal Length Distribution by Species")
plt.xlabel("Petal Length")
plt.ylabel("Density")

plt.show()
```

Output:



Step 6: Contour Plot (2D Density)

```
plt.figure(figsize=(8,6))

sns.kdeplot(
  data=df,
  x='petal length (cm)',
  y='petal width (cm)',
  fill=True
)

plt.title("Contour Plot (Petal Length vs Width)")
plt.xlabel("Petal Length")
plt.ylabel("Petal Width")

plt.show()
```

Output: